

SCI225 Pathophysiology — Midterm High-Yield Review

Units 1–6 • Chapters 1, 2, 4, 6, 9, 11, 14, 16–20

Study Spine — ask 3 questions about every finding:

1. What physiologic change is happening? 2. What mechanism caused it? 3. Is there a compensatory response?

MECHANISM (how it happens)

WHY (cause → effect)

COMPENSATION (body's response)

PATHOLOGY (clinical result)

Unit 1 • Cellular Biology, Genetics, Adaptation & Injury Ch 1, 2, 4

1.1 Cell Biology — Fast Reference

Organelle	Function	Transport	Key idea
Nucleus	DNA storage; transcription	Passive diffusion	Down gradient, no ATP
Mitochondria	ATP via oxidative phosphorylation; apoptosis trigger	Facilitated diffusion	Via protein channel, no ATP
Ribosomes	Protein synthesis (translation)	Active transport	Against gradient, needs ATP (Na ⁺ /K ⁺ pump)
Rough ER	Protein synthesis/folding	Osmosis	Water moves to higher solute
Smooth ER	Lipid synthesis, detox, Ca ²⁺ storage	Endo/exocytosis	Bulk transport with vesicles
Golgi	Modify/package proteins		
Lysosome	Intracellular digestion		
Peroxisome	Detox reactive species		

MECHANISM DNA → *transcription* → mRNA → *translation (ribosome)* → Protein. Disrupt any step → abnormal or absent protein → disease.

1.2 Genetics — Mutation, Inheritance, Chromosomes

Pattern	Key features	Example
Autosomal Dominant	One copy → disease; affects every generation; 50% offspring risk	Huntington, Marfan, familial hypercholesterolemia
Autosomal Recessive	Both copies needed; carriers unaffected; skips generations	Cystic fibrosis, sickle cell, PKU, Tay-Sachs
X-linked Recessive	Males affected; mother carrier; no male-to-male transmission	Hemophilia A, Duchenne MD, color blindness
Multifactorial	Genes + environment	HTN, DM2, neural tube defects

Chromosomal (Nondisjunction)	Karyotype	Hallmarks
Down syndrome	Trisomy 21	Intellectual disability, flat facies, cardiac defects
Turner syndrome	45, XO	Short stature, webbed neck, ovarian failure (female)
Klinefelter	47, XXY	Tall, hypogonadism, gynecomastia (male)
Cri-du-chat	5p deletion	High-pitched cry, microcephaly

WHY Cystic fibrosis — CFTR gene defect → impaired Cl^- transport across epithelium → water doesn't follow → thick mucus → airway obstruction, pancreatic insufficiency, infertility.

MECHANISM Epigenetics: heritable changes *without* altering DNA sequence. **DNA methylation** silences genes; **histone modification** (acetylation) loosens chromatin → more expression. Key in cancer and development.

1.3 Cellular Adaptation — 5 Types

Type	Change	Trigger	Example
Atrophy	↓ cell size	Disuse, denervation, ↓ hormone/nutrients	Casted limb muscle
Hypertrophy	↑ cell size	↑ workload	Cardiac muscle in HTN; skeletal muscle in athletes
Hyperplasia	↑ cell number	Hormonal or compensatory	Endometrium in pregnancy; liver regeneration
Metaplasia	One adult cell type → another (reversible)	Chronic irritation	Smoker's bronchus: ciliated columnar → squamous
Dysplasia	Disordered growth, size/shape variation	Chronic injury → pre-cancer	Cervical dysplasia (HPV)

1.4 Cell Injury, Necrosis & Apoptosis

Hypoxic injury cascade

↓ O_2 → ↓ ATP → Na^+/K^+ pump fails → $\text{Na}^+ + \text{H}_2\text{O}$ into cell → **cell swelling** (reversible) → Ca^{2+} influx → enzyme activation (phospholipases, proteases) → membrane damage → **necrosis** (irreversible)

Necrosis subtypes		Necrosis	Apoptosis
Coagulative	Solid organ ischemia (MI, kidney)	Unregulated	Regulated, programmed
Liquefactive	Brain infarct, abscess	Cell swells/bursts	Cell shrinks into bodies
Caseous	TB granuloma — "cheese-like"	Inflammation yes	Inflammation no
Fat	Pancreatitis, trauma to breast	Pathologic	Physiologic or pathologic
Fibrinoid	Vessel walls (vasculitis, malignant HTN)		
Gangrenous	Limb ischemia (dry/wet)		

MECHANISM Apoptosis pathways: Intrinsic — mitochondrial; cytochrome c release → caspase cascade (DNA damage, withdrawal of growth factors). Extrinsic — death receptors (Fas/TNF) → caspase cascade.

WHY Free radicals (ROS: O_2^- , H_2O_2 , $OH^\bullet$) damage membranes via lipid peroxidation, denature proteins, break DNA. Defenses: SOD, catalase, glutathione, vitamins C/E. Overwhelm → ischemia-reperfusion injury, aging, cancer.

1.5 Practice Questions

Q1. A smoker's biopsy shows replacement of ciliated columnar epithelium with stratified squamous cells. This is:

A. Atrophy B. Hyperplasia C. Metaplasia D. Dysplasia

[Show answer & rationale](#)

C. Metaplasia. One mature cell type replaced by another to tolerate chronic stress. Reversible if stimulus removed; can progress to dysplasia/cancer. A = size ↓. B = number ↑. D = disordered (pre-malignant) but cells don't change type.

Q2. In hypoxic injury, cell swelling occurs because:

A. ATP ↑ forces water in B. Na^+/K^+ pump fails → Na^+ and water accumulate C. Membrane permeability ↓ D. Ca^{2+} efflux

[Show answer](#)

B. No ATP → pump stops → intracellular Na^+ rises → water follows osmotically → swelling. This is still *reversible* until membrane damage.

Q3 (SATA). Which are consistent with apoptosis (not necrosis)? Select all.

A. Cell shrinkage B. Surrounding inflammation C. Caspase activation D. Membrane rupture E. Programmed, ATP-dependent

[Show answer](#)

A, C, E. Apoptosis is orderly, non-inflammatory, caspase-mediated. B and D describe necrosis.

Q4. A child has Trisomy 21. The most likely cause is:

A. Autosomal recessive inheritance B. Nondisjunction during meiosis C. X-linked mutation D. Mitochondrial inheritance

[Show answer](#)

B. Failure of chromosome pairs to separate during meiosis → gamete with extra chromosome → Trisomy 21.

Unit 2 • Immunity, Inflammation, Infection Ch 6, 7, 8, 9

2.1 Innate Immunity (nonspecific, immediate)

Line	Components	Role
1st	Skin, mucous membranes, cilia, normal flora, pH, lysozyme in tears	Physical/chemical barrier — keep pathogens out
2nd	Phagocytes (neutrophils, macrophages), NK cells, complement, inflammation, fever, cytokines	Rapid nonspecific response; no memory

MECHANISM Complement (classical/alternative/lectin) → opsonization (C3b), chemotaxis (C5a), membrane attack complex (C5b-9 → lyse microbe), inflammation.

2.2 Adaptive Immunity (specific, slower, has memory)

	Humoral (B cells)	Cell-mediated (T cells)
Target	Extracellular pathogens, toxins	Intracellular pathogens, tumor, transplant
Effector	Plasma cells → antibodies	CD4 helper, CD8 cytotoxic
MHC	MHC II presents to CD4	MHC I presents to CD8

Ig	Key role
IgG	Most abundant; crosses placenta; secondary response
IgM	Largest; first produced; primary response
IgA	Mucosal surfaces, breast milk
IgE	Allergy/parasites; binds mast cells
IgD	B-cell receptor (activation)

WHY Active immunity = you make your own antibodies (infection or vaccine); long-lasting. **Passive** = you receive them (placenta, breast milk, Ig injection); temporary.

2.3 Acute Inflammation & Wound Healing

Acute inflammation cascade

Injury → mast cell degranulation → histamine, prostaglandins, bradykinin → vasodilation (redness, warmth) + ↑ permeability (edema) → leukocyte margination → diapedesis → chemotaxis → phagocytosis → resolution or chronic inflammation

Cardinal sign	Mechanism
Redness (rubor)	Vasodilation ↑ blood flow
Heat (calor)	↑ blood flow
Swelling (tumor)	↑ capillary permeability → fluid/protein leak
Pain (dolor)	Bradykinin, prostaglandins, pressure on nerves
Loss of function	Swelling + pain

Wound healing phase	What happens	Timing
Hemostasis	Vasoconstriction, clot forms	Min–hrs
Inflammatory	Neutrophils → macrophages; debride	0–3 days
Proliferative	Fibroblasts, collagen, angiogenesis, granulation	3–21 days
Remodeling	Collagen reorganizes; scar matures	Wks–yrs

PATHOLOGY Dysfunctional healing: keloid (excess collagen), contracture (scar shortening limits motion), dehiscence (wound separation), evisceration (organ protrusion).

2.4 Hypersensitivity — All Four Types

Type	Mediator	Timing	Examples
I — Immediate	IgE on mast cells → histamine	Min	Anaphylaxis, asthma, hay fever, food allergy
II — Cytotoxic	IgG/IgM vs cell-surface antigen	Min–hrs	Hemolytic transfusion reaction, Graves, myasthenia gravis, Goodpasture
III — Immune complex	Ag-Ab complexes deposit	Hrs	SLE, serum sickness, post-strep GN, RA
IV — Delayed/T-cell	Sensitized T cells (no antibody)	2–3 days	TB skin test (PPD), contact dermatitis (poison ivy), transplant rejection

MECHANISM Anaphylaxis (Type I): allergen cross-links IgE on mast cells → massive histamine release → systemic vasodilation (↓ BP), ↑ permeability (edema), bronchoconstriction → hypotension + airway obstruction. Treat: epinephrine.

2.5 Autoimmune & Immunodeficiency

Autoimmune	Target
SLE	Anti-dsDNA, ANA — multisystem (skin, kidneys, joints)
Rheumatoid arthritis	Synovium → joint destruction (pannus)
Myasthenia gravis	Anti-ACh receptor at NMJ (Type II)
Multiple sclerosis	CNS myelin (Type IV)
Hashimoto / Graves	Thyroid (see Unit 6)
Type 1 DM	β-cell destruction

Immunodeficiency	Defect
Primary (congenital)	Genetic — SCID (B+T), Bruton's (B cells), DiGeorge (thymus/T cells)
Secondary (acquired)	HIV (CD4 destruction), chemo, steroids, malnutrition, aging

WHY HIV targets CD4 helper T cells → progressive loss → impaired coordination of both humoral and cell-mediated immunity → opportunistic infections at CD4 < 200 (AIDS).

2.6 Infection & Stress Response

Infection sequence

Encounter → entry → colonization → invasion → multiplication → tissue damage (toxins, host inflammation)

MECHANISM Fever: pyrogens (LPS, IL-1, TNF, IL-6) → hypothalamus raises thermoregulatory set point → shivering/vasoconstriction until new set point reached. Benefits: slows bacterial growth, enhances immune function.

WHY Stress response (Selye GAS): Alarm (sympathetic + epi) → Resistance (HPA axis: CRH → ACTH → cortisol; ↑ glucose, ↓ inflammation, ↓ immune) → Exhaustion (if chronic → ↑ infection, HTN, DM, depression).

2.7 Practice Questions

Q5. A bee sting causes hypotension, hives, and wheezing within minutes. This is mediated by:

A. IgG complement fixation B. IgE on mast cells releasing histamine C. T-cell cytotoxicity D. Immune complex deposition

[Show answer](#)

B. Type I hypersensitivity/anaphylaxis — IgE cross-linking triggers mast cell degranulation → histamine → vasodilation, permeability, bronchoconstriction.

Q6 (SATA). Early acute inflammation involves:

A. Vasodilation B. ↑ capillary permeability C. Neutrophil migration D. Memory B-cell proliferation E. Histamine release

[Show answer](#)

A, B, C, E. D is adaptive immunity — not the early driver of acute inflammation.

Q7. A patient receives a TB skin test and develops induration at 48 hours. This is:

A. Type I B. Type II C. Type III D. Type IV

[Show answer](#)

D. Type IV delayed T-cell hypersensitivity. No antibodies involved; sensitized T cells react 24–72 hours after exposure.

Q8 (cloze). In anaphylaxis, the allergen binds ____ on mast cells; the mediator released is ____.

[Show answer](#)

IgE ; histamine.

Unit 3 • Oncology Ch 11, 12, 13

3.1 Benign vs Malignant

	Benign	Malignant
Growth	Slow, encapsulated	Rapid, invasive
Differentiation	Well differentiated (resemble origin)	Poorly differentiated / anaplastic
Metastasis	No	Yes
Recurrence	Rare after removal	Common
Naming	tissue + "-oma" (lipoma, adenoma)	carcinoma (epithelial), sarcoma (connective), leukemia/lymphoma (blood)

3.2 Hallmarks of Cancer & Key Genes

- Self-sufficient proliferation signaling (oncogenes)
- Insensitivity to anti-growth signals (tumor suppressor loss)
- Evasion of apoptosis
- Limitless replicative potential (telomerase)
- Sustained angiogenesis
- Invasion & metastasis
- Altered metabolism (Warburg) & immune evasion

Gene type	Normal role	In cancer	Examples
Oncogene	Drives cell division (accelerator)	Gain-of-function mutation → constant "on"	RAS, MYC, HER2, BCR-ABL
Tumor suppressor	Brakes division, repairs DNA, triggers apoptosis	Loss-of-function → no brakes	p53 ("guardian of genome"), RB, BRCA1/2, APC
DNA repair genes	Fix errors	Loss → mutations accumulate	MLH1, MSH2 (Lynch)

WHY Angiogenesis — tumors >1–2 mm need new vessels (VEGF-driven) to receive nutrients/O₂; without it, cancer cells can't grow or spread.

3.3 Spread, Staging, Systemic Effects

Metastasis

Local invasion → intravasation (enter vessel/lymph) → circulation → extravasation → seed new organ → angiogenesis at new site

Grading	Staging (TNM)
How abnormal the cells look (differentiation). G1 = well differentiated; G4 = anaplastic.	How far spread. T tumor size, N nodes, M metastasis. Stage 0 = in situ; Stage IV = distant spread.

PATHOLOGY Cachexia — progressive loss of muscle/fat despite intake. Driven by TNF, IL-6, tumor metabolism. Worsens prognosis.

PATHOLOGY Paraneoplastic syndromes — remote effects from tumor-secreted substances: SIADH (small-cell lung), hypercalcemia (squamous lung, PTHrP), Cushing (ACTH-secreting), polycythemia (RCC).

Carcinogens	Example cancer
Tobacco	Lung, bladder, oral
UV radiation	Skin (melanoma, SCC, BCC)
HPV	Cervical, oropharyngeal
Hepatitis B/C	Hepatocellular
EBV	Burkitt, nasopharyngeal
H. pylori	Gastric
Asbestos	Mesothelioma

3.4 Practice Questions

Q9. A tumor biopsy shows overexpression of growth factor receptors and continuous proliferative signaling. Which mechanism is active?

A. Tumor suppressor gain of function B. Oncogene activation C. Restored differentiation D. Enhanced apoptosis

[Show answer](#)

B. Oncogenes are "gain-of-function" — activated drive continuous growth signals. Tumor suppressors go the other way (loss).

Q10. A small tumor cannot grow past 2 mm without:

A. Apoptosis B. Angiogenesis C. Differentiation D. Cell cycle arrest

[Show answer](#)

B. Angiogenesis — new vessels (VEGF) supply O₂ and nutrients; diffusion alone supports only very small tumors.

Q11. TNM staging assesses:

A. Cell differentiation grade B. Tumor size, node involvement, metastasis C. Mutation burden D. Apoptotic index

[Show answer](#)

B. Staging = how far it has spread; grading = how abnormal cells look.

Unit 4 • Structure & Function of the Nervous System Ch 14

4.1 CNS / PNS & the Neuron

CNS	PNS	Neuron part	Role
Brain + spinal cord	Cranial nn, spinal nn, ganglia	Dendrites	Receive input
Oligodendrocytes myelinate	Schwann cells myelinate	Soma	Cell body
No regen (mostly)	Can regenerate	Axon	Conducts AP away
		Myelin / nodes of Ranvier	Saltatory conduction
		Axon terminal	Release NT

Brain region	Function
Frontal	Executive, motor, personality, Broca's (speech prod.)
Parietal	Somatosensory
Temporal	Auditory, memory, Wernicke's (speech comp.)
Occipital	Vision
Cerebellum	Balance, coordination
Brainstem	Cardio-respiratory centers, CN nuclei
Hypothalamus	Homeostasis, hormones, temperature

4.2 Action Potentials & Neurotransmission

Action potential

Resting (-70 mV) → stimulus opens Na^+ channels → depolarization → threshold reached → Na^+ rushes in (spike) → K^+ out (repolarization) → hyperpolarization → Na^+/K^+ pump restores

MECHANISM Saltatory conduction: AP jumps node → node along myelinated axon → faster, energy-efficient. Demyelination (MS, Guillain-Barré) slows conduction → weakness, numbness.

Neurotransmitter	Effect / Clinical link
Acetylcholine (ACh)	NMJ, parasympathetic; ↓ in Alzheimer's, blocked in myasthenia
Dopamine	Reward, motor control; ↓ Parkinson's, ↑ schizophrenia
Serotonin	Mood, sleep; ↓ depression
Norepinephrine	Arousal, sympathetic
GABA	Main inhibitory; ↓ seizures
Glutamate	Main excitatory; excitotoxicity in stroke
Endorphins	Endogenous pain suppression

Reflex arc (monosynaptic)

Stimulus → receptor → sensory (afferent) neuron → spinal cord → motor (efferent) neuron → effector muscle → contraction

4.3 CSF, BBB & Cerebral Circulation

MECHANISM CSF produced by choroid plexus of ventricles → flows through lateral → 3rd → 4th → subarachnoid space → absorbed by arachnoid villi. Cushions brain, delivers nutrients, removes waste.

MECHANISM Blood-Brain Barrier: tight endothelial junctions + astrocyte foot processes. Selectively permeable — excludes most pathogens/drugs; allows O₂, CO₂, glucose, lipid-soluble molecules.

WHY Cerebral autoregulation maintains flow across MAP 60–160 mmHg. Below → ischemia; above → edema/hemorrhage. Brain uses ~20% of cardiac output despite being 2% of body weight — very vulnerable to hypoxia.

4.4 Practice Questions

Q12. In MS, slowed impulse conduction is due to:

- A. Axon severance B. Disrupted saltatory conduction from demyelination C. Blocked synaptic vesicles
D. ACh receptor blockade

[Show answer](#)

B. MS = autoimmune destruction of CNS myelin → loss of saltatory conduction → slow/incomplete APs.

Q13 (order). Arrange the reflex arc:

1. Motor neuron 2. Receptor detects stimulus 3. Sensory neuron 4. Muscle contracts

[Show answer](#)

2 → 3 → 1 → 4.

Unit 5 • Disorders of the Nervous System Ch 16, 17, 18

5.1 Increased ICP, Stroke

MECHANISM **Monro-Kellie:** skull is rigid with 3 fixed contents (brain 80%, blood 10%, CSF 10%). If one ↑, another must ↓ or ICP rises. Normal ICP < 15 mmHg.

Increased ICP cascade

↑ ICP → ↓ cerebral perfusion pressure (CPP = MAP - ICP) → ischemia → edema → further ↑ ICP → herniation → death

PATHOLOGY **Cushing's triad** (late, ominous sign of ↑ ICP): HTN (widened pulse pressure) + bradycardia + irregular respirations. Reflects brainstem compression.

	Ischemic stroke (~85%)	Hemorrhagic stroke (~15%)
Cause	Thrombus or embolus blocks artery	Vessel rupture (HTN, aneurysm, AVM)
Mechanism	↓ blood flow → ↓ ATP → Na ⁺ /K ⁺ fail → glutamate excitotoxicity → neuronal death	Blood compresses tissue + ↑ ICP
Onset	Sudden focal deficit	Sudden worst-headache, ↓ LOC
Tx window	tPA ≤ 4.5 hr if no bleed	Control BP, surgery; tPA contraindicated

WHY **Why focal deficits in stroke:** specific artery territory loses O₂ → neurons in that region fail → symptoms map to function (MCA → contralateral face/arm weakness, aphasia if left hemisphere).

5.2 TBI & Seizures

TBI type	Description
Concussion	Transient altered consciousness, no structural damage on imaging
Contusion	Bruise; coup (impact site) and/or contrecoup (opposite side)
DAI (diffuse axonal injury)	Shearing of white matter from rotational forces → severe, often coma
Epidural hematoma	Arterial (middle meningeal); "lucid interval" then deterioration
Subdural hematoma	Venous (bridging veins); slower, common in elderly/alcoholics
Subarachnoid hemorrhage	Ruptured aneurysm; "thunderclap" headache

Seizure type	Features
Focal (partial)	One hemisphere; simple (conscious) or complex (impaired)
Generalized tonic-clonic	LOC, stiffening (tonic) then jerking (clonic), postictal
Absence	Brief staring spells (children)
Status epilepticus	> 5 min or repeated without recovery — emergency

MECHANISM Seizure = abnormal, excessive synchronous neuronal firing. Imbalance of glutamate (↑ excitatory) vs GABA (↓ inhibitory). Causes: idiopathic, trauma, stroke, tumor, electrolyte, fever, withdrawal.

5.3 Degenerative & Demyelinating Disorders

Disorder	Pathophysiology	Hallmarks
Alzheimer's	Amyloid plaques + tau tangles; ↓ ACh	Progressive memory loss, cortical atrophy
Parkinson's	Loss of dopaminergic neurons in substantia nigra	Resting tremor, rigidity, bradykinesia, postural instability
Huntington's	AD — CAG trinucleotide repeat → atrophy of caudate/putamen	Chorea, dementia, psychiatric
Multiple Sclerosis	Autoimmune demyelination CNS (Type IV)	Relapsing-remitting; optic neuritis, weakness, paresthesias
ALS	Both upper + lower motor neuron degeneration	Progressive weakness, atrophy, fasciculations; cognition spared

5.4 NMJ, PNS, Meningitis, Spinal Cord

Disorder	Mechanism	Clinical
Myasthenia gravis	Anti-ACh receptor antibodies (Type II)	Fatigable weakness, worse with use, ocular first (ptosis, diplopia)
Guillain-Barré	Autoimmune demyelination of PNS (often post-infection)	Ascending paralysis, areflexia, risk of respiratory failure
Bacterial meningitis	Inflammation of meninges (S. pneumo, N. meningitidis)	Fever, stiff neck, photophobia, ↓ LOC; ↑ WBC/↑ protein/↓ glucose CSF
Viral meningitis	Enteroviruses	Milder; normal glucose

SCI level	Result
C1–C4	Tetraplegia + loss of diaphragm (ventilator)
C5–C8	Tetraplegia, some arm function
T1–T12	Paraplegia, trunk/leg loss
Lumbosacral	Variable leg/bladder/bowel

PATHOLOGY Spinal shock — immediate areflexia and flaccid paralysis below injury; resolves over weeks. **Autonomic dysreflexia** (injury above T6) — stimulus below injury → massive SNS → severe HTN, bradycardia, headache. **Emergency:** sit upright, find trigger (often bladder distention).

5.5 Pediatric Neurology

Condition	Key point
Hydrocephalus	CSF accumulation → ↑ head circumference, bulging fontanel, sunset eyes. Sutures not yet fused allow expansion.
Neural tube defects	Failure of neural tube closure; folate deficiency. Spina bifida, meningocele, myelomeningocele, anencephaly.
Cerebral palsy	Non-progressive motor disorder from prenatal/perinatal brain injury; spastic most common.
Febrile seizures	6 mo–5 yr; rapid temp rise; usually benign.
Reye syndrome	Encephalopathy + liver failure after viral illness + aspirin in children.

5.6 Practice Questions

Q14 (case). 68-year-old with sudden right-sided weakness + slurred speech, CT no bleed. Mechanism?

A. CSF excess compressing cortex B. Autoimmune demyelination C. ↓ cerebral blood flow → neuronal hypoxia D. Peripheral motor neuron degeneration

[Show answer](#)

C. Ischemic stroke. ↓ flow → ↓ ATP → ion pump failure → glutamate excitotoxicity → focal neuronal death.

Q15. 24-year-old with weakness that worsens through the day, improves with rest, sensation intact. Mechanism?

A. CNS demyelination B. Basal ganglia dopamine loss C. Impaired ACh receptor function at NMJ D. Cortical ischemia

[Show answer](#)

C. Myasthenia gravis. Anti-ACh receptor antibodies → fatigable weakness, ocular first. Sensation intact.

Q16 (matrix). Match mechanism to finding:

1. ↑ vascular permeability → ? 2. ↓ cerebral flow → ? 3. ACh receptor blockade → ? 4. Angiogenesis → ?

[Show answer](#)

1—Edema, 2—Focal neuro deficit, 3—Fatigable weakness, 4—Tumor growth support.

Q17. Infant with rapidly enlarging head + bulging fontanel. Cause?

A. CSF accumulating in ventricles B. NMJ antibody block C. PNS demyelination D. Synaptic vesicle failure

[Show answer](#)

A. Hydrocephalus. Unfused sutures allow cranial expansion.

Unit 6 • Endocrine System Ch 19, 20

6.1 Hormonal Regulation Principles

Concept	Key point
Negative feedback	High hormone inhibits its own stimulus (maintains homeostasis)
Positive feedback	Amplifies (oxytocin in labor, LH surge)
Water-soluble	Peptides/catecholamines → bind surface receptors → 2nd messenger (cAMP) → fast
Lipid-soluble	Steroids/thyroid → enter cell → intracellular/nuclear receptors → alter gene transcription → slower, longer
Up-regulation	Low hormone → ↑ receptors → ↑ sensitivity
Down-regulation	High hormone → ↓ receptors → ↓ sensitivity (e.g., insulin resistance in T2DM)

HPT axis (example)

Low T3/T4 → Hypothalamus TRH ↑ → Pituitary TSH ↑ → Thyroid T3/T4 ↑ → Negative feedback shuts off TRH/TSH

6.2 Pituitary — SIADH vs Diabetes Insipidus

	SIADH (too much ADH)	DI (too little ADH or renal resistance)
Mechanism	↑ water reabsorption	↓ water reabsorption
Serum Na ⁺	Low (dilutional hyponatremia)	High
Urine	Concentrated, low volume	Dilute, huge volumes (polyuria)
Symptoms	Water intoxication, HA, seizures	Polydipsia, dehydration
Causes	Small-cell lung Ca, CNS injury, meds	Central (pituitary) or nephrogenic

PATHOLOGY Acromegaly / Gigantism — GH excess (usually adenoma). Before growth plates close → gigantism; after → acromegaly (enlarged hands/feet/jaw, organomegaly, insulin resistance).

6.3 Thyroid & Parathyroid

	Hyperthyroidism (Graves)	Hypothyroidism (Hashimoto)
Mechanism	TSI antibody stimulates TSH receptor (Type II)	Autoimmune destruction of thyroid
TSH / T4	TSH ↓, T4 ↑	TSH ↑, T4 ↓ (primary)
Metabolism	↑ — weight loss, heat intoler., tachy, tremor, diarrhea, exophthalmos	↓ — weight gain, cold intoler., brady, fatigue, constipation, dry skin
Crisis	Thyroid storm — fever, tachyarrhythmia, delirium	Myxedema coma — hypothermia, ↓ LOC, hypoventilation

Parathyroid	Effect
Hyperparathyroidism	↑ PTH → ↑ Ca ²⁺ (bones, stones, groans, psychiatric overtones)
Hypoparathyroidism	↓ PTH → ↓ Ca ²⁺ → tetany, Chvostek/Trousseau signs, seizures

6.4 Adrenal Disorders

	Cushing (↑ cortisol)	Addison (↓ all adrenal hormones)
Cause	Pituitary adenoma (ACTH ↑), adrenal tumor, exogenous steroids	Autoimmune destruction of cortex (primary)
Cortisol	↑	↓
BP / glucose	HTN, hyperglycemia	Hypotension, hypoglycemia
Appearance	Moon face, buffalo hump, central obesity, striae, easy bruising	Hyperpigmentation (↑ AChT), weight loss, fatigue
Electrolytes	↑ Na ⁺ , ↓ K ⁺ (some aldosterone effect)	↓ Na ⁺ , ↑ K ⁺ (loss of aldosterone)
Crisis	—	Addisonian crisis — shock, hypoglycemia

PATHOLOGY Pheochromocytoma — catecholamine-secreting adrenal medulla tumor → episodic HTN, HA, palpitations, diaphoresis ("5 P's").

6.5 Diabetes Mellitus

	Type 1	Type 2
Mechanism	Autoimmune β-cell destruction → absolute insulin deficiency	Insulin resistance + relative deficiency (receptor down-regulation)
Onset	Child/young adult, abrupt	Adult, gradual (obesity link)
Body habitus	Thin	Often obese
Ketosis	Prone (DKA)	Resistant (HHS more typical)
Treatment	Insulin (always)	Lifestyle, oral agents, eventually insulin

WHY Classic 3 P's (both types): *polyuria* (glucose osmotic diuresis), *polydipsia* (dehydration), *polyphagia* (cells starved despite high glucose).

	DKA	HHS
Who	T1 (usually)	T2 (usually)
Glucose	> 250	> 600
Ketones	Present (fat breakdown → ketoacids)	Minimal (residual insulin blocks ketogenesis)
pH	< 7.3 (metabolic acidosis)	Normal
Signs	Kussmaul breathing, fruity breath, N/V, abd pain	Severe dehydration, ↓ LOC, focal deficits
Tx	IV fluids, insulin, K ⁺ replacement	IV fluids (massive), insulin

PATHOLOGY Chronic DM complications: micro (retinopathy, nephropathy, neuropathy) + macro (CAD, stroke, PAD). Driven by glycation end-products + vascular damage.

6.6 Practice Questions

Q18. A patient exposed to chronically elevated hormone levels will most likely show:

- A. Up-regulation + ↑ sensitivity B. Down-regulation + ↓ sensitivity C. Permanent loss of 2nd messenger
D. Loss of carrier proteins

[Show answer](#)

B. Persistent excess → ↓ receptor number/affinity → insulin resistance is a classic example.

Q19 (cloze). When T3/T4 fall, hypothalamus releases _____ and anterior pituitary releases _____.

[Show answer](#)

TRH ; TSH.

Q20. A patient with weight gain, moon face, hyperglycemia, and HTN most likely has:

- A. Addison's B. Cushing's C. Hashimoto's D. Pheochromocytoma

[Show answer](#)

B. Cushing's. Cortisol excess → central obesity, moon face, hyperglycemia, HTN, immunosuppression.

Q21. A T1 diabetic presents with Kussmaul breathing, fruity breath, glucose 420, pH 7.15. Mechanism?

- A. HHS — severe dehydration without ketosis B. DKA — absolute insulin deficiency → fat breakdown → ketoacids C. Hypoglycemia D. SIADH

[Show answer](#)

B. DKA. No insulin → cells can't use glucose → lipolysis → ketones → metabolic acidosis → Kussmaul compensatory hyperventilation.

Q22 (SATA). Which findings fit primary Addison's disease?

- A. Hyperpigmentation B. Hypertension C. Hyponatremia D. Hyperkalemia E. Moon face

[Show answer](#)

A, C, D. Loss of cortisol + aldosterone → ↓ BP, ↓ Na⁺, ↑ K⁺. ↑ ACTH drives pigmentation. B + E = Cushing.

Appendix • High-Yield "Why" Statements & Compensation Summary

Every "Why" You Need

Finding	Why it happens
Edema	↑ capillary permeability OR ↑ hydrostatic pressure OR ↓ oncotic pressure → fluid into interstitium
Fever	Pyrogens (IL-1, TNF, IL-6) reset hypothalamic set point upward
Cell swelling in hypoxia	↓ ATP → Na ⁺ /K ⁺ pump fails → Na ⁺ + water accumulate intracellularly
Cancer metastasis	Loss of adhesion + enzymatic invasion + angiogenesis + vascular entry + distant seeding
Stroke focal deficits	Specific arterial territory loses O ₂ → specific function fails
Hormone target specificity	Only cells with matching receptor respond
Down-regulation	Chronic high hormone → fewer/less sensitive receptors
Insulin resistance (T2DM)	Chronic hyperinsulinemia → receptor down-regulation + post-receptor defects
Myasthenia weakness that worsens with use	ACh receptors already reduced; repeated stimulation depletes remaining signaling
Demyelination slows conduction	No myelin = no saltatory jumping → AP travels continuously (slow)
Cushing's triad in ↑ ICP	Brainstem compression → SNS surge (HTN) → baroreflex bradycardia + irregular respirations
Kussmaul breathing in DKA	Respiratory compensation blows off CO ₂ to offset metabolic acidosis
Polyuria in DM	Glucose in filtrate exceeds reabsorption → osmotic diuresis
Hyperpigmentation in Addison's	Low cortisol → ↑ ACTH → ACTH precursor stimulates melanocytes
Anaphylactic hypotension	Massive histamine → systemic vasodilation + plasma leak → ↓ preload/afterload

Compensation Cheat Sheet

Stressor	Compensation
↓ BP (hemorrhage, anaphylaxis)	SNS: ↑ HR, vasoconstriction; RAAS: Na ⁺ /water retention
↓ O ₂ (hypoxia)	↑ RR, ↑ HR, ↑ EPO → RBC production
Metabolic acidosis	Kussmaul breathing (↓ CO ₂), renal ↑ HCO ₃ ⁻ reabsorption
↑ ICP	Displace CSF, then venous blood; eventually fails → herniation
↓ Thyroid hormone	TRH ↑ → TSH ↑ (tries to stimulate gland)
↑ Hormone chronic	Receptor down-regulation
Infection	Fever + ↑ WBC + inflammation + adaptive immunity
Tissue injury	Vasodilation, permeability, leukocytes, healing phases

Final Study Tips

- **Explain out loud.** If you can say *why* each finding happens, you're ready.
- **Follow the cascade.** Most questions ask you to trace stimulus → mechanism → result → compensation.
- **Watch for triads.** Cushing's triad (ICP), 3 P's (DM), classic anaphylaxis triad (hypotension + airway + skin).
- **Differentiate look-alikes.** DKA vs HHS; Cushing vs Addison; Hashimoto vs Graves; SIADH vs DI; ischemic vs hemorrhagic stroke.
- **Up vs down.** Low hormone → up-regulation ↑ sensitivity; high hormone → down-regulation ↓ sensitivity.

SCI225 Pathophysiology Midterm High-Yield Review — built from your Unit 1–6 review, expanded to fill content gaps in adaptive immunity, autoimmune disease, specific cancers, neurologic disorders, endocrine diseases including diabetes, Cushing/Addison, SIADH/DI, thyroid, and pediatric neuro.